



Techno-economic analysis of energy micro-grids with hydrogen storage and fuel cell in Moroccan farming systems

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ARTICLE INFO

Editor name: Mohamed Fathy El-Amin Mousa

Keywords:

Sustainable farming

Microgrids

Rural electrification

Hydrogen-powered systems

Techno-economic analysis

ABSTRACT

This study evaluates the techno-economic performance of hybrid renewable microgrids integrating hydrogen storage and fuel cells in two Moroccan pilot farms: a grid-connected site (BLFARM) and an off-grid site (RIMSAR). Real meteorological and load data were analyzed in HOMER Pro to assess feasibility. In 2024, BLFARM achieved a Levelized Cost of Energy (LCOE) of €1.63/kWh and a Renewable Fraction (Ren Frac) of 83.9%, while RIMSAR reached €4.32/kWh with 100% renewable contribution. Hydrogen use remained limited due to low demand and high costs. Assuming 2050 hydrogen-technology reductions, LCOE decreased to €0.160/kWh (BLFARM) and €0.425/kWh (RIMSAR), while hydrogen components were still underutilized. Aggregating demand from 5-80 farms reduced LCOE by over 50% from €0.093 to €0.045/kWh (BLFARM) and from €0.142 to €0.074/kWh (RIMSAR) while increasing electrolyzer and fuel-cell operation. Community-networked hydrogen microgrids thus enhance component utilization, energy resilience, and cost effectiveness in rural Moroccan agriculture.

Introduction

Global food security challenges are driving modern agricultural practices to maximize productivity, leading to increased energy and water demands [1]. This increase mainly impacts rural farming communities with limited access to power grids, making energy supply and resource management challenging. Renewable energy systems (RESs) provide promising solutions for agricultural power needs, reducing greenhouse gas emissions and operational costs (OCs). By reducing reliance on fossil fuels, these systems can enhance the sustainability of farming operations, especially for energy-intensive tasks like water pumping and irrigation.

Despite the potential of renewable energy, many Moroccan farmers continue to rely on costly and environmentally damaging non-renewable energy sources, such as diesel-powered pumps, primarily due to limited storage options for renewable energy and the lack of grid connectivity in rural areas [2]. In fact, the intermittent nature and production uncertainty of RES require efficient storage systems to capture excess energy, ensuring its availability during periods of no production, especially, for farms lacking access to public grids.

Solar photovoltaic (PV) systems in farming often use lead-acid batteries, which are not well-suited for water pumping, especially in hot regions, where high temperatures accelerate aging and degrade their state of health. The storage of electricity as hydrogen has gained significant interest in recent decades due to its cleanliness and efficiency.

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<https://doi.org/10.1016/j.sciaf.2025.e03087>

Received 12 August 2025; Received in revised form 9 October 2025; Accepted 8 November 2025

Available online 12 November 2025

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List of Abbreviations	
Abbreviation	Description
AC	Alternating Current
API	Application Programming Interface
BESS	Battery Energy Storage Systems
BLFARM	Grid-connected farm (specific project)
CRF	Capital Recovery Factor
DC	Direct Current
EMS	Energy Management System
FC	Fuel Cell
HES	Hydrogen Energy Systems
HHM	Hydrogen-enhanced Hybrid Microgrid
HOMER	Hybrid Optimization of Multiple Energy Resources (software)
IoT	Internet of Things
IRENA	International Renewable Energy Agency
LCOE	Levelized Cost of Energy
LEAP-RE	Long-Term Joint EU–AU Research and Innovation Partnership on Renewable Energy
LF	Load Following
LPG	Liquefied Petroleum Gas
MESRSI	Ministère de l'Enseignement Supérieur, de la Recherche Scientifique et de l'Innovation
MGFARM	Smart Stand-Alone Microgrids for Agricultural Electrification (project)
NPC	Net Present Cost
OC	Operational Cost
O&M	Operation and Maintenance
PEM	Proton Exchange Membrane (e.g., PEM electrolyzer, PEM fuel cell)
PEMFC	Proton Exchange Membrane Fuel Cell
PV	Photovoltaic
Ref	Reference
Ren Frac	Renewable Fraction
RES	Renewable Energy System
RIMSAR	Off-grid farm (specific project)
SOFC	Solid Oxide Fuel Cell
WT	Wind Turbine

Among emerging storage technologies, hydrogen is receiving considerable interest as a carbon-free energy carrier that can store energy for long periods of time and is flexible in its applications [3]. It can be produced through several routes, including thermochemical, biological, and electrochemical processes. The most promising pathway for sustainable applications is water electrolysis using renewable energy sources. This is referred to as green hydrogen, since it supports carbon-free production and utilization cycles [4]. The main electrolysis technologies include alkaline, proton exchange membrane (PEM), and solid oxide electrolyzers, each differing in operating temperature, efficiency, and cost [4,5]. Once produced, hydrogen can be stored, transported, and then converted back to electricity with fuel cells. This hydrogen creates a more sustainable option to balance the variability of renewable generation. PEM fuel cells are of particular interest because of their high conversion efficiency, low emissions, and fast dynamic response, making them a promising technology for microgrids and rural applications [6]. Recent techno-economic studies further emphasize the role of hybrid renewable hydrogen microgrids, where coupling photovoltaics or wind with electrolyzers and storage systems enhances energy autonomy and reduces lifecycle costs [5].

Hydrogen storage might play a pivotal role in farming energy systems and generally in supporting rural communities [7,8]. However, further techno-economic studies need to be conducted in order to explore the role hydrogen storage might play in enhancing the sustainability and resilience of farming energy systems. Particularly, within Moroccan context, studies must be performed to further shed more light on how hydrogen can ensure stable energy supply for agricultural operations, e.g., irrigation, farming processing activities, refrigeration, farm machinery and mobility. These studies aim to highlight how hydrogen integration can foster energy independence while creating new economic opportunities for farmers, which will contribute to a more sustainable and self-sufficient agricultural sector, towards energy, water and food nexus [2]. It is worth noting that, over the past decade, considerable work has been done on hybrid renewable energy systems for rural electrification, increasingly focusing on choosing

energy storage options and the economic feasibility of implementation. However, to the best of our knowledge, studies concerning the deployment of these hybrid systems in Moroccan context while using actual farming data remain limited.

The work conducted in this paper is part of the MGFARM¹ project (Smart Stand-Alone Microgrids for Agricultural Electrification) where the main objective is to develop Smart Micro-Grids based on the production of renewable energy in order to support sustainable development of smart and energy-efficient farming systems [1,2]. The project aims to (i) deploy digitized microgrids and smart storage solutions, (ii) develop distributed approaches, alternatives for decentralized energy management, by interconnecting microgrids in which farms can share excess energy, sell it to neighboring farms or store it for later use, ensuring energy efficiency while enhancing sustainability [9], and (iii) incorporate a blockchain-based IoT platform for secure, decentralized energy sharing and efficient resource management within interconnected microgrids [10].

The present research work investigates the potential of hydrogen-enhanced hybrid microgrids (HHMs) for farming systems. A HHM typically combines electrolyzers, storage tanks, and fuel cells (FCs) to convert surplus energy from wind and solar sources into hydrogen for later use. We mainly examine two representative case studies in Morocco, BLFARM and RIMSAR farms, which serve as innovative testbeds for integrating sustainable energy solutions into agricultural operations through photovoltaic (PV) systems, yet continue to face limitations that hinder the full deployment of advanced renewable solutions. More precisely, the work focuses on developing HHM configurations tailored to the electricity needs of these Moroccan farms, alongside conducting simulations and techno-economic evaluations of hybrid microgrid systems using HOMER (Hybrid Optimization of Multiple Energy Resources), complemented by MATLAB Simulink-based modeling. The analysis is further extended to 2050 to assess how advancements in hydrogen component costs and performance could influence system feasibility, while also exploring community-scale applications to enhance the broader relevance of the findings beyond individual farms.

In summary, the primary contributions and originality of this research are as follows:

- Establish a comprehensive feasibility assessment framework for HHMs dedicated to agricultural applications, combining technical, economic, and environmental evaluation under Moroccan operating conditions.
- Pioneer a dual-scale modeling framework analyzing two representative farms: BLFARM (grid-connected) and RIMSAR (off-grid), using real agricultural load and on-site meteorological data to quantify hydrogen's contribution to autonomy and cost reduction.
- Extend the investigation to future-scenario and community-scale studies up to the year 2050, projecting the impact of declining hydrogen technology costs and scaling the results to clustered farming communities.

The remainder of this paper is structured into six main sections to provide a clear overview of the research and its outcomes. Following the introduction that establishes the context and objectives, Section “Related work” presents existing works from the literature, with a focus on techno-economic studies and assessments. Section “Renewable Energy Systems: An Overview” provides an overview of renewable energy systems, emphasizing hybrid configurations, sizing, and modeling. Section “Materials and Methods” describes the materials and methods. Section “Simulation Results” analyzes the simulation results. Finally, Section “Conclusions and Future Work” concludes the study and outlines future perspectives.

Related work

Early studies essentially looked at grid-connected setups. For example, Caliskan and Percin [11] present an on-grid hydrogen-producing scheme for a university campus, which proved economically feasible but was applicable in only a few instances to off-grid agricultural domains. Likewise, Mohammad and Iqbal [12] studied solar-hydrogen microgrid networks in urban settings, demonstrating the cost-effectiveness of integrating FCs but not particularly addressing agricultural electrification.

In contrast, off-grid solutions have increasingly been explored for rural development. For instance, Köprü et al. [13] primarily assessed a stand-alone hybrid microgrid for rural electrification, pairing hydrogen with batteries to achieve greater energy autonomy. This research concentrated on general access in rural areas and not on the on-farm requirements. A notable example is the case study by Bagdadee and Zhang [14], which evaluated a hybrid microgrid integrating PV, wind, micro-hydro, diesel, and storage in Kuakata, a coastal rural area of Bangladesh. Their research established the technical, economic, and environmental feasibility of hybrid systems for rural electrification in off-grid communities. Mainly, the focus was on general household and community electrification. Dawood et al. [15] designed an off-grid microgrid run by 100% renewable energy source incorporating solar PV, battery, and hydrogen storage, demonstrating potential for HHMs, but was based upon theoretical load assumptions rather than actual data. Abdolmaleki and Berardi [16] studied solar-hydrogen hybrid systems for residential buildings, where the long-term storage potential of hydrogen was proved. Nonetheless, their results were based on modeled climate data, limiting the actual use of the findings. Agyekum et al. [17] optimized hydrogen integrated hybrid energy systems using a multi-criteria decision-making process, but no real load data was considered, which makes practical validation difficult.

Studies on microgrid optimization have also developed strategies for the design and integration of microgrid systems. Mandić et al. [18] looked into standalone microgrids with hydrogen and battery storage in relation to urban resilience, without linking the results to agriculture. Emezirinwune et al. [19] studied hybrid renewable systems for rural development in Nigeria, which show the relation between energy access and agricultural productivity. Their study emphasized the importance of decentralized microgrids, but the study did not incorporate hydrogen storage, which lowered the reliability of the system. Rehman et al. [4] extended this

¹ MGFARM: Smart Stand-Alone Microgrids for Agricultural Electrification

line of research to a large-scale wind-hydrogen system in the Aqaba Gulf, Saudi Arabia, where HOMER simulations showed that wind farms coupled with electrolyzers and hydrogen storage could reliably meet residential and industrial demand, with feasibility strongly influenced by wind variability and hydrogen costs.

In the agricultural sector, renewable microgrid research has focused almost entirely on irrigation and energy access. For example, Coelho et al. [20] compared the models HOMER and HY4RES in irrigation energy solutions, showing that hybrid microgrids can effectively reduce water pumping costs. Yet their study did not incorporate hydrogen storage and thus was not prolonged over long-term energy autonomy. In addition, Ross-Hopley et al. [21] analyzed renewable energy integration into irrigation projects in Saskatchewan using HOMER Pro, showing that even partial inclusion of solar and wind significantly reduced irrigation energy costs. On a similar note, Tiong et al. [22] examined the microgrid resilience with hydrogen storage, but their work emphasized stability of the system, not optimization of agricultural energy demands.

There are other studies that have evaluated microgrids that operate independently for farming. For instance, Olatunji et al. [23] cited off-grid microgrids for farm electrification, establishing feasibility but excluding hydrogen-based storages. Xia et al. [24] analyzed Hydrogen Enhanced Plant accepting stability with renewable fluctuations, rather than agricultural needs, and coalesced their attention on home applications. In a recent study by Farhani et al. [25], a solar-hydrogen hybrid system was put forth for an agricultural farm in Kairouan, Tunisia. Their model is fundamentally different from the existing one in that it does not employ battery storage to store energy; instead, hydrogen serves that function through electrolysis. Also, it includes a unit of water desalination directed toward irrigation water, thus demonstrating water-energy nexus approaches in agriculture. However, their analysis was limited to a stand-alone configuration and did not explore on-grid integration or community-scale farm aggregation, which are key aspects addressed in the current study.

Similarly, Alturki et al. [26] explored the design of a hybrid PV-Wind-Diesel-Battery-Hydrogen system to serve an industrial city in Saudi Arabia and was able to achieve high renewable fractions with green hydrogen, but these results are useful to an agricultural operation that would be completely different to a large scale energy supply. Akarsu et al. [27] developed an optimal hybrid PV-Wind-Diesel-Battery-Hydrogen system to supply residential demand in Kayseri, Turkey, while achieving 68% renewable penetration and without sacrificing hydrogen storage for greater energy reliability, but their work focused on urban scale residential loads instead of agriculture-specific microgrids.

From the survey we conducted, it is clear that although hybrid renewable systems have been vastly explored, techno-economic studies, which specifically integrate HHM for agricultural applications, remain widely unexplored, particularly at the community level. This study aims to bridge this gap by examining both on on-grid and off grid small-scale HHM for farming, considering their deployment at individual farms as well as within community-networked configurations. A detailed comparison of prior studies is presented in Table 1, which highlights their methodological and technical attributes across the literature. It showcases the scope of each study in regard to grid connection types, whether or not it involved the use of real load and on-site meteorological data, including battery, hydrogen storage, and fuel cells, and whether agricultural context and community-scale farms were presented. Notably, most existing works focus on general rural electrification, while very few address the specific energy needs of farms, especially within the Moroccan context or on a community-scale.

In summary, it is evident that while hybrid renewable systems have been extensively explored for rural and residential electrification, techno-economic studies focusing on HHMs for agriculture applications remain scarce, particularly at the community scale. In Morocco, these systems are still at an early stage of development, and currently, existing agricultural studies rarely consider hydrogen-based storage or fuel-cell integration using real load and meteorological data. The work presented in this paper aims to bridge this gap by evaluating the feasibility and optimal sizing of HHMs for two Moroccan farms, one grid-connected and one off-grid chosen to be a representative energy system for rural areas. Mainly, this study can be regarded as one of the first integrated techno-economic evaluations of hydrogen-enhanced agricultural microgrids in Morocco, addressing real-world data and national priorities on green hydrogen and sustainable farming, while offering a replicable framework for similar regions across North Africa.

Renewable energy systems: An overview

This section introduces RESs together with required components that are required to deploy hybrid energy systems, we are considering in the present study. For instance, solar PV systems generate electricity by converting sunlight directly into electrical power through the PV effect, where semiconductors generate an electric current upon exposure to sunlight. Known for their environmental benefits, PV systems significantly reduce greenhouse gas emissions and reliance on fossil fuels. Continuous advancements in PV materials, such as monocrystalline and polycrystalline silicon and emerging perovskite-based cells, have enhanced efficiency and affordability. However, PV systems face variability due to factors like irradiance, temperature, and angle of installation, while producing energy only during daylight hours. Thus, integrating PV with energy storage or complementary sources, like wind, improves system stability and ensures a continuous energy supply [28]. HOMER Pro computes the power output of a PV array based on the following equation [29] (1), where Y_{PV} denotes the power output under specified test conditions (kW), whereas f_{PV} represents the PV derating factor (%). The parameter $\overline{G_T}$ corresponds to the solar radiation incident on the PV array at the given time step (kW/m²), while $G_{T,STC}$ signifies the incoming radiation under standard test conditions (1 kW/m²). The coefficient α_P defines the power temperature coefficient, expressed in (%/°C). Furthermore, T_c indicates the PV cell temperature at the current time step (°C), whereas $T_{c,STC}$ designates the PV cell temperature under standard test conditions, typically 25 °C [29].

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{\overline{G_T}}{G_{T,STC}} \right) [1 + \alpha_P (T_c - T_{c,STC})] \quad (1)$$

Table 1

Comparison of the proposed method to prior research works.

Reference	HOMER	On-grid	Off-grid	Real load data	Real meteorological data	Battery storage	H ₂ storage	Fuel Cell based	Agricultural application	Community farms
[4]	✓	✓	✗	✓	✓	✗	✓	✗	✗	✗
[11]	✓	✓	✗	✗	✓	✗	✓	✗	✗	✗
[12]	✓	✓	✓	✓	✓	✓	✗	✗	✗	✗
[13]	✓	✗	✓	✓	✓	✓	✓	✓	✗	✗
[14]	✓	✗	✓	✓	✓	✓	✗	✗	✗	✗
[15]	✓	✗	✓	✗	✓	✓	✓	✓	✗	✗
[16]	✓	✗	✓	✓	✓	✓	✓	✓	✗	✗
[17]	✓	✗	✓	✗	✓	✓	✓	✗	✓	✗
[18]	✓	✗	✓	✗	✓	✓	✓	✓	✗	✗
[19]	✓	✗	✓	✓	✓	✓	✗	✗	✓	✗
[20]	✓	✓	✗	✗	✓	✓	✗	✗	✓	✗
[21]	✓	✓	✓	✓	✓	✓	✗	✗	✓	✗
[22]	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
[23]	✓	✗	✓	✗	✓	✓	✗	✗	✓	✗
[24]	✓	✗	✓	✗	✓	✓	✓	✗	✗	✗
[25]	✓	✗	✓	✓	✓	✗	✓	✓	✓	✗
[26]	✓	✗	✓	✗	✓	✓	✗	✗	✗	✗
[27]	✓	✗	✓	✗	✓	✓	✓	✗	✗	✗
The proposed study	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Unlike PVs, wind turbines (WTs) capture wind's kinetic energy to produce electricity, making wind one of the most developed and economically viable renewable energy technologies. Effective in high wind areas, WTs provide complementary power to solar PVs, particularly during non-solar hours. Modern turbine designs incorporate advanced materials and aerodynamics to enhance energy capture and resilience in harsh conditions. However, wind energy's variability necessitates integration with energy storage or hybridization with other renewable sources, like PV, to stabilize the supply and improve grid reliability [30]. The wind speed is given in Eq. (2), where U_{hub} denotes the wind speed measured at the hub height of the WT (m/s), while U_{anem} represents the wind speed recorded at the anemometer height (m/s). The variable z_{hub} corresponds to the hub height of the WT (m), whereas z_{anem} signifies the anemometer height (m). Additionally, Z_0 characterizes the surface roughness length (m) [29].

$$U_{\text{hub}} = U_{\text{anem}} \frac{\ln\left(\frac{z_{\text{hub}}}{Z_0}\right)}{\ln\left(\frac{z_{\text{anem}}}{Z_0}\right)} \quad (2)$$

Battery Energy Storage Systems (BESS), are vital for addressing renewable energy fluctuations by storing excess electricity generated during peak production and releasing it when demand is high or renewable output is low. BESS enable grid stability, smoothing energy supply fluctuations and allowing for greater renewable penetration. When combined with PV and WT, BESS reduces reliance on fossil fuel backup systems, enhancing overall system reliability [31,32]. The amount of power $P_b(t)$ (kW) that the BESS can store is calculated according to Eq. (3), where Q_1 denotes the amount of energy available in the batteries (kWh), while the parameter Q corresponds to the total amount of energy (kWh), whereas c defines the storage capacity ratio. Additionally, k represents the storage rate constant, and Δt indicates the time interval (hours) [29].

$$P_b(t) = \frac{kQ_1(t)e^{-k} + Q(t)kc(1 - e^{-k\Delta t})}{1 - e^{-k\Delta t} + c(k\Delta t - 1 + e^{-k\Delta t})} \quad (3)$$

The BESS's maximum power discharge capacity within a certain time frame $P_b(t)$ (kW) is determined using Eq. (4), where Q_{max} represents the total storage capacity (kWh).

$$P_b(t) = \frac{-kQ_{\text{max}} + kQ_1(t)e^{-k\Delta t} + Q(t)kc(1 - e^{-k\Delta t})}{1 - e^{-k\Delta t} + c(k\Delta t - 1 + e^{-k\Delta t})} \quad (4)$$

Converters and inverters are critical for hybrid systems with both Direct Current (DC) and Alternative Current (AC) components, such as PV, WT, and BESS [17]. Converters are used to invert DC electricity produced by PV panels or stored in the battery into AC to supply electrical loads, as well as to rectify any excess AC energy produced by WTs into DC to either power the electrolyzer or charge the batteries [33]. The converter output is always less than converter input due to inherent energy losses during conversion, in the HOMER model this is quantified as a conversion efficiency rate typically below 100%. The input and output power relationship is expressed as follows [34]:

$$P_{\text{out}}^{\text{Con}} = P_{\text{in}}^{\text{Con}} \times \eta_{\text{con}} \quad (5)$$

where η_{con} is the efficiency of the converter (%), where $P_{\text{in}}^{\text{Con}}$ and $P_{\text{out}}^{\text{Con}}$ are the input and output power of the converter, respectively, expressed in kilowatts (kW).

Similarly, electrolyzers convert excess renewable electricity into hydrogen through electrolysis, splitting water into hydrogen and oxygen. Hydrogen serves as a versatile energy carrier, stored for later conversion back to electricity [13]. The electrolyzer used in the proposed study is a PEM electrolyzer. As a DC device, it is either directly interfaced with DC buses or connected via DC-DC power converters to operate at its rated voltage level [35]. PEM water electrolyzers are chosen over Alkaline water electrolyzers for their greater operational flexibility, longer lifespan, higher efficiency, greater operational flexibility, and simpler design [36]. The electrochemical reactions governing the process are outlined in Eqs. (6) (7) (8) [13,37,38]:



FC generates electrical energy from chemical energy through electrochemical reactions. Different types of FCs exist, including proton exchange membrane fuel cells (PEMFCs), solid oxide fuel cells (SOFC), and molten carbonate FCs. Among them, the PEMFC is often preferred for grid-connected applications and standalone power systems due to its high power density, low operating temperature and zero emissions. However, despite these advantages, it also has drawbacks, including a slow response to transient loads [12]. In a FC, electricity is generated through the combination of hydrogen and oxygen. Hydrogen is fed into the anode, while oxygen or air is supplied to the cathode. The electrochemical reactions responsible for this process are outlined in the following Eqs. (9) (10) (11) [13]:



It is worth noting that the FC is modeled as a generator that utilizes hydrogen as its fuel type. Therefore, key parameters include details on the fuel curve, emission levels, maintenance requirements, and operational schedule [12]. The fuel consumption rate F_c (kg/h) is expressed by Eq. (12), where f_o is the generator fuel curve intercept coefficient (kg/h/kW), Y_{gen} is the rated capacity of the generator (kW), f_i is the generator fuel curve slope (kg/h/kW), and P_{gen} represents the electrical output of the generator (kW) [29].

$$F_c = f_o Y_{gen} + f_i P_{gen} \quad (12)$$

Hydrogen storage is a long-term storage solution, which complements BESS by providing a backup when renewable production is low or BESS capacity is exceeded [39]. If hydrogen production is unavailable for a prolonged period or if there is an oversupply, the hydrogen tank functions as a storage solution, ensuring a continuous supply while improving system reliability without requiring replacement over the project's lifespan [11]. The Hydrogen Energy System (HES) includes the electrolyzer, the FC and the hydrogen storage tank.

The utility grid constitutes a large-scale interconnected framework that enables the generation, transmission, and distribution of electrical energy across vast geographical areas, thereby guaranteeing reliability and efficiency in the provision of energy [40]. This main grid acts as an additional backup to ensure energy availability when renewable sources cannot meet demand [41]. The grid power price in Morocco is considered in our study. In addition, propane Liquefied Petroleum Gas (LPG) generators could be integrated into the system, with cost parameters, which are derived from market prices of LPG generators in Morocco [42–44].

Materials and methods

The BLFARM and RIMSAR farms in Morocco serve as models for integrating sustainable energy into agriculture to efficiently meet energy needs. The study collects real-time meteorological data (solar irradiance, wind speed, temperature). This data is transmitted to an IoT platform for storage, preprocessing, and validation [45]. HOMER Pro was selected for simulating, designing, and optimizing the hybrid microgrid system. It enables comprehensive modeling of hybrid configurations by integrating data on renewable resources, load demand, and component characteristics. Its simulation capabilities allow for iterative adjustments, testing different configurations, and analyzing economic feasibility, including cost-benefit outcomes and sensitivity analyses [46,47]. The software was utilized to model and assess the feasibility and performance of HHMs for the two small-scale pilot farms: BLFARM and RIMSAR. It is worth noting that HOMER is a widely adopted software platform for modeling both conventional and renewable energy systems [48]. It supports simulation, optimization, and sensitivity analysis for off-grid and grid-connected microgrids, allowing researchers to design and assess hybrid configurations by defining the optimal power capacities of the system's components, while also performing techno-economic evaluations to determine the most cost-effective and reliable solutions [48,49]. Featuring an extensive library of technologies and components, HOMER can run complex simulations of energy systems, delivering optimized system designs that align with the given load profile and site conditions [48,50].

The proposed methodological framework of the techno-economic analysis is illustrated in Fig. 1. It begins with defining input data, including load profiles and meteorological parameters (solar irradiance, wind speed, and temperature), followed by selecting system components with their technical and cost specifications. HOMER is then used to simulate and optimize system's configurations through an iterative process until a feasible solution is reached. Finally, technical, economic, and environmental analyses are performed to determine the optimal system sizing.

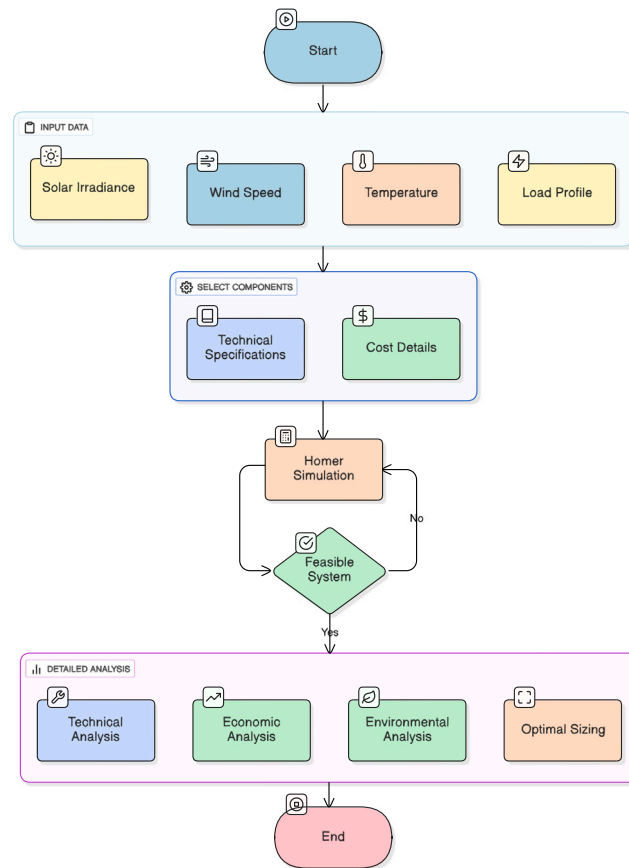


Fig. 1. Flowchart of the techno-economic analysis methodology used in this study.

Sites description

The BLFARM and RIMSAR farms, showcase innovative models for integrating sustainable energy into agriculture. BLFARM, spanning 5 hectares in Saniat Berguig, Sidi Bennour Province, within the Casablanca-Settat region, continues to rely heavily on the national grid to sustain irrigation and other energy-intensive operations (see Fig. 2(a)). Despite taking into consideration the implementation of a photovoltaic system, the farm has yet to achieve significant renewable energy penetration. This dependence on conventional electricity leads to high operational costs and increased vulnerability to grid instabilities, especially during peak agricultural demand periods (e.g., night irrigation and water pumping). On the other hand, RIMSAR, located in Oulad Sbih, El Kalaa des Sraghna, covers 12 hectares and operates as an off-grid farm (see Fig. 2(b)). Even with a PV system in place for irrigation, the farm continues to face challenges in maintaining a consistent energy supply, especially during periods of low solar irradiance, and consequently relies on fuel-based generators. The limited scale of its hybrid renewable system restricts its ability to support agricultural growth and fully satisfy the energy requirements.

Taken together, the two sites illustrate the persistent barriers to renewable energy deployment in Moroccan farming systems. BLFARM illustrates the ongoing reliance on grid electricity, while RIMSAR highlights the limitations of small standalone off-grid systems, particularly regarding reliability and scalability. These challenges justify the need for more robust hybrid microgrid solutions, including advanced storage options, such as hydrogen, to ensure resilience, cost-effectiveness, and sustainable agricultural productivity.

Data collection and pre-processing

The data collection process was carefully designed to ensure continuous and accurate real-time monitoring of all variables required for optimizing the renewable-powered irrigation system. At both pilot sites, BLFARM and RIMSAR, meteorological data, including solar irradiance, wind speed, and ambient temperature, are measured directly on site using calibrated sensors. The collected data are sampled at fixed time intervals and averaged to create daily and seasonal profiles. These profiles help HOMER simulator capture the natural variations in renewable energy availability and maintain an appropriate balance between energy supply and irrigation demand.



Fig. 2. Geographic location: (a) BLFARM (b) RIMSAR.

All data from the sensors, including environmental readings and irrigation loads, are transmitted to the *MGFARM* platform through the MQTT protocol [1]. Basic pre-processing is performed on the IoT layer to filter out errors and ensure data consistency before analysis. Data transmission follows the same secure MQTT-based path from field devices to the edge node, as outlined in our previous work [51].

To make the data available for analysis and simulation, the *MGFARM* platform connects to a Python application through an Application Programming Interface (API). This interface enables smooth data exchange, allowing the application to access the latest measurements, such as hourly solar irradiance, wind speed, temperature, and irrigation loads, and use them to generate daily and seasonal averages for the energy simulation.

The data are then cleaned to remove errors caused by sensor noise, power cuts, or communication interruptions. After cleaning, a validation step checks the consistency and reliability of the remaining data to ensure that only realistic values are kept. Once validated, all variables are resampled to an hourly resolution, which is the time scale required by the energy system simulator. This resampling step also helps identify missing periods or gaps in the data. Custom routines automatically detect these missing timestamps and generate a “gap map” that highlights incomplete hours, days, or months [52].

Each missing period is recorded in a summary log for traceability. Short gaps are filled using interpolation methods that take into account the daily behavior of the variables, for example by using data from the same hour on nearby days. Longer missing periods are filled using representative values taken from similar seasonal conditions or from regional climate references. Throughout this process, the original sensor readings are always preserved, and any added synthetic data remain minimal to keep the results realistic. Finally, a light smoothing step removes abrupt changes created during the filling process and produces a continuous, coherent dataset ready for hybrid energy and irrigation simulations.

This workflow ensures that the final datasets are both realistic and consistent with real farm conditions, while remaining fully compatible with the *MGFARM* framework and the secure communication path defined in our earlier works.

Load profiles

Detailed load profiles were developed for each site to capture the unique energy demands associated with agricultural activities. These profiles reflect the power requirements for irrigation, water pumping, and other critical farm operations throughout the day and across different seasons. The load data were acquired through on-site monitoring and historical records of energy consumption. This high-resolution data provides a realistic basis for sizing system components and selecting storage options to meet peak demand reliably.

For BLFARM, load data was collected directly from on-site sensors. In contrast, RIMSAR only had load data available on the existing residential components, as the irrigation pumps were installed at a later stage. To address this gap, MATLAB Simulink was used to estimate the electrical power, which is required by the pumping system. The simulated values were then used as the representative load profile of the irrigation system.

In this study, the irrigation pumping system model consists of a motor, which is mechanically coupled to a centrifugal pump (see Fig. 3). The motor transforms electrical energy into mechanical torque to drive the pump, to deliver hydraulic power to the irrigation system. The nominal hydraulic power of the pumping system $P_{h \text{ nominal}}$ installed on the farm is 5 kW. The system was modeled in MATLAB Simulink (see Fig. 4), based on sensor-derived inputs, such as solar irradiation and temperature. Since the motor is the final electrical component in the pumping system, our focus is placed on calculating the electrical power delivered from the inverter to the motor. This value captures the actual energy consumption of the system and serves as the primary input for defining the irrigation load profile. When the PV array output exceeds the power requirement of the pump motor, the extra energy is lost and does not contribute to pumping water. To account for this, the electrical power, supplied to the pump at each time step t , is calculated accordingly. The electrical power required to operate the pump, denoted as $P_{\text{pump input}}(t)$ (kW), is determined by

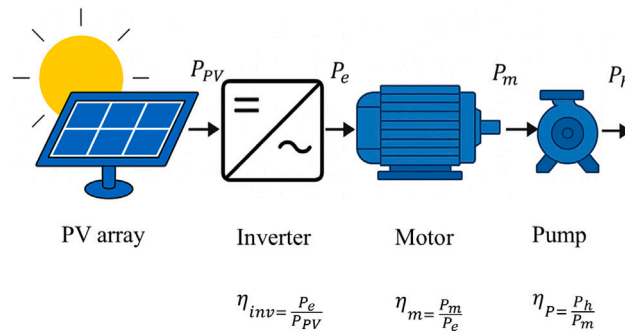


Fig. 3. Power flow in the PV-powered pumping system.

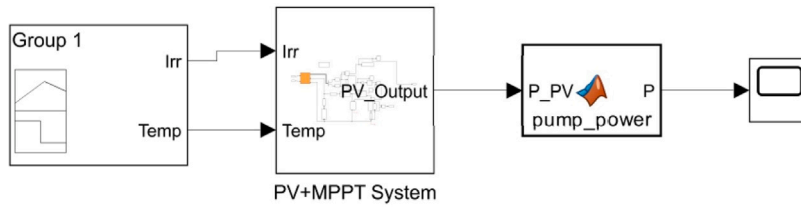


Fig. 4. Simulink model for pump power calculation.

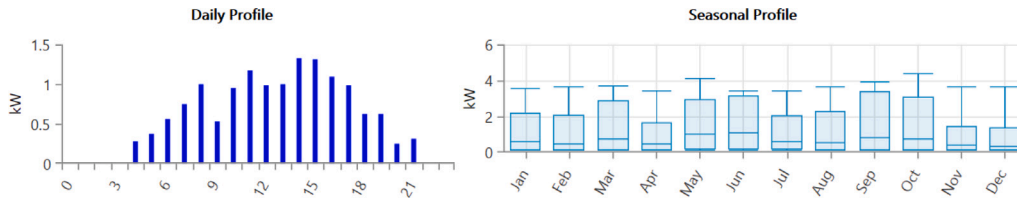


Fig. 5. BLFARM Load profile.

selecting the minimum between the available electrical power from the PV system and the nominal electrical input limit of the pump. This approach ensures that the pump never receives more power than it is rated for [53,54]:

$$P_{\text{pump input}}(t) = \min(P_{\text{elec}}(t), P_{\text{elec nominal}}) \quad (13)$$

The instantaneous electrical power delivered by the PV system, $P_{\text{elec}}(t)$ (kW), accounts for the power output of the PV generator and the efficiency losses in the inverter and the motor driving the pump. It is calculated as follows, where $P_{\text{PV output}}(t)$ represents the PV array output power at time t (kW), η_{inv} is the inverter efficiency (%), and η_{motor} is the motor efficiency (%).

$$P_{\text{elec}}(t) = P_{\text{PV output}}(t) \cdot \eta_{\text{inv}} \cdot \eta_{\text{motor}} \quad (14)$$

The nominal electrical input power of the pump, $P_{\text{elec nominal}}$ (kW), is derived from the rated hydraulic power of the pump, $P_{\text{h nominal}}$ (kW), and the pump efficiency η_{pump} (%), as follows:

$$P_{\text{elec nominal}} = \frac{P_{\text{h nominal}}}{\eta_{\text{pump}}} \quad (15)$$

Eqs. (13)–(15) were implemented in a MATLAB/Simulink function block to compute the pump's hourly electrical demand, which was then exported as a load profile. To calculate the actual electrical input required, it was necessary to account for conversion losses across the inverter, motor, and pump. The efficiency values used were:

$$\eta_{\text{inv}} = 95\%, \eta_{\text{motor}} = 80\%, \eta_{\text{pump}} = 85\% \quad [55].$$

The implemented function determines the electrical power input to the pump at each time step by comparing the available electrical power from the PV system with the pump's rated electrical capacity. First, the PV electrical output is calculated by multiplying the PV power output with the inverter and motor efficiencies. This represents the maximum usable power at the pump's input. The function then takes the minimum between this value and the pump's nominal electrical power derived from the rated hydraulic power divided by the pump efficiency.

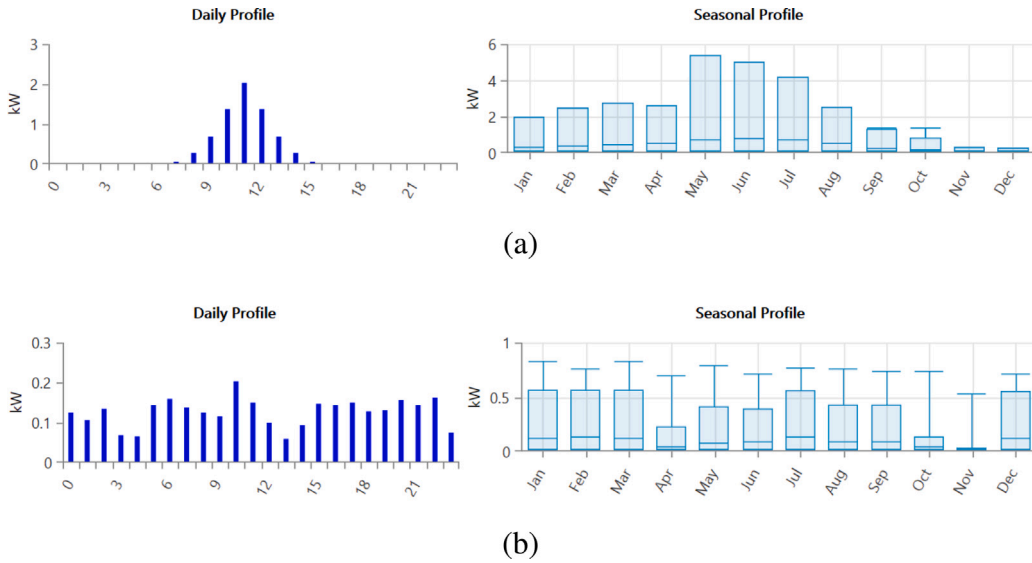


Fig. 6. RIMSAR Load Profile: (a) Irrigation system (b) the remaining farm's loads.

In this study, a simplified power estimation method was adopted instead of modeling a full power electronics system. This approach was chosen to reduce simulation complexity and computational time, which is especially important for long-term techno-economic analysis. The goal was to estimate energy flows and determine optimal system's sizing rather than simulating fast electrical transients. To further simplify the simulation, the PV input data (irradiation and temperature) were averaged over a representative day for each month. This allowed the generation of one daily PV power profile per month, which was then extended to represent the entire month. This method was repeated for all 12 months to create a yearly energy profile that is computationally efficient yet sufficiently representative for the study's objectives. The MATLAB-derived irrigation load was exported in CSV format and directly imported into HOMER Pro simulations as demand inputs.

Fig. 5 shows the daily loads and seasonal load profiles for the BLFARM site. The daily profile shows a single pronounced peak during the central hours of the day, with a smooth curve, indicating that energy consumption primarily concentrates around typical hours of operation. The seasonal profiles indicates fairly steady demand throughout the year, with mild increases during some months, likely indicating seasonal agricultural activities. These profiles indicate mixed use that incorporate both residential and farm energy usage.

Fig. 6 displays the disaggregated electrical load profiles of the RIMSAR site. Fig. 6(a) corresponds to the irrigation load profile obtained from MATLAB simulation of the pumping system. It shows a sharply peaked daily profile, with significant seasonal variation, with the largest demand during spring and summer, consistent with increased water requirements from crops during the growing season. Fig. 6(b) represents the measured residential and auxiliary farm loads obtained from on-site sensors. This load exhibits a more dispersed daily profile without the distinct peak and seasonal variation observed in the irrigation demand. The total RIMSAR load therefore combined the measured residential demand with the simulated irrigation demand generated in MATLAB.

It should be emphasized that MATLAB was employed exclusively for generating the irrigation load profiles, whereas the techno-economic modeling and system optimization were conducted entirely within HOMER Pro.

Evaluation metrics

The cost analysis applied a multi-criteria methodology, integrating economic, technical, and social dimensions. Metrics, such as Net Present Cost (NPC), Levelized Cost of Energy (LCOE), Operating Cost, and Initial Capital emphasized financial feasibility, while technical parameters like Operation & Maintenance (O&M) costs and Renewable Fraction (Ren Frac) highlighted system performance [13,56].

For instance, the LCOE is a standard metric for comparing electricity generation technologies and assessing economic feasibility [57]. The LCOE (€/kWh) is calculated using the following equation, where E_t denotes the total electrical load served (kWh/yr), while C_{annual} represents the total annualized cost of the system (€/kWh) [29,58]:

$$LCOE = \frac{C_{\text{annual}}}{E_t} \quad (16)$$

NPC (€) accounts for the total sum of costs and revenues accumulated throughout the system's lifetime. The NPC (€) is calculated using the following expression:

$$NPC = \frac{C_{\text{annual}}}{CRF_{i,R_{\text{proj}}}} \quad (17)$$

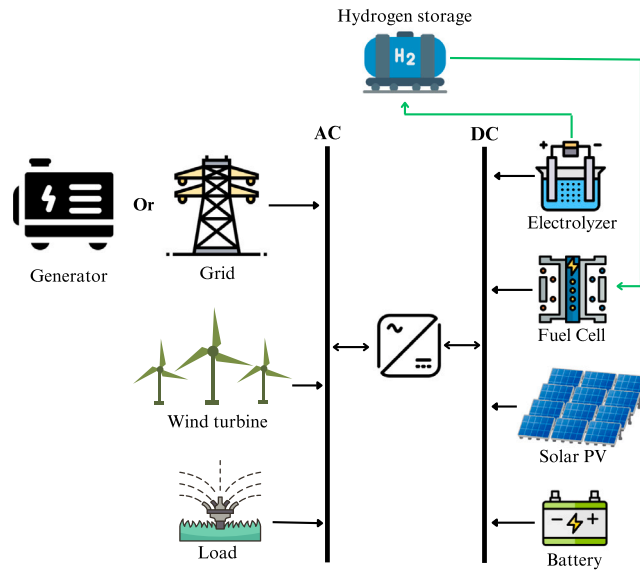


Fig. 7. Proposed hybrid systems.

The real interest rate is represented by i (%), while R_{proj} indicates the system's lifetime (yr), and CRF is the capital recovery factor, n is number of years.

$$CRF(i,n) = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (18)$$

The real discount rate i is determined using the following, where i^* (%) is the nominal discount rate, and F is the expected inflation rate (%).

$$i = \frac{i^* - F}{1 + F} \quad (19)$$

The discount factor f_d is a ratio used to assess the present value of a cash flow that happens at any time during the course of project. HOMER calculates this factor using the following formula:

$$f_d = \frac{1}{(1+i)^n} \quad (20)$$

System's design and assessment

In this study, as stated above, we examine two distinct agricultural sites: one grid-connected and one islanded. BLFARM, the grid-connected farm, combines its existing PV system with access to the main electricity grid. In contrast, RIMSAR, the islanded farm, operates independently of the main grid and is currently equipped with a PV system alongside a propane LPG generator. This study proposes extending these farms to hybrid energy systems, as depicted in Fig. 7, incorporating PV, WT, BESS, and a HES, which includes an electrolyzer, FC, and a hydrogen storage tank, as well as either a grid connection or an LPG generator, adapted to the specific needs of each farm.

Fig. 8 illustrates the system architecture modeled for the two pilot farms: (a) BLFARM and (b) RIMSAR. In the HOMER simulations, a controller was applied to manage power distribution and integrate energy sources. For this study, the Load Following (LF) dispatch strategy was adopted to optimize efficiency by ensuring generators meet real-time demand, reducing excess generation and fuel consumption. This strategy prioritizes renewable energy sources, aims to meet load demand as long as possible, and minimizes system's outages. Batteries are refilled with renewable energy, and the generator operates at minimum load levels when necessary [13].

Table 2 summarizes the technical and economic parameters of all components for both farms. These parameters were implemented directly in HOMER, together with the meteorological data, the full hourly load data measured at the BLFARM site, and the RIMSAR demand described in Section "Load profiles", which combines measured residential loads with the simulated irrigation load generated in MATLAB.

For the community-scale scenarios, aggregated demand was approximated by multiplying the single-farm load profiles of BLFARM and RIMSAR by the number of farms (ranging from 5 to 80). This approach was applied to both the grid-connected community and the islanded community cases. The resulting community load profiles were then imported into HOMER for optimization [63].

Table 2
Economic parameters of the components.

Farms	Component	Size	Unit	Initial Capital Cost (€/kW)	Replacement Cost (€/kW)	O&M (€/year)	Lifetime (year)	Ref
BLFARM	Solar PV module	3.84	kW	112	0	1.12	25	[56]
	Wind Turbine	3	kW	1690	1690	17	20	[59]
	Electrolyzer	1	kW	1620	1620	50	15	[60]
	Hydrogen tank	1	kg	600	600	3	25	[60]
	Fuel Cell	1	kW	370	370	0.019	50 000 (hour)	[61]
	Battery	2	Quantity	218.40	218.40	2.18	–	[56]
	Grid	0.1	€/kW	–	–	–	–	[62]
	Converter	1	kW	375	375	10	25	[59]
RIMSAR	Solar PV module	17.16	kW	112	0	1.12	25	[56]
	Wind Turbine	3	kW	1690	1690	17	20	[59]
	Electrolyzer	1	kW	1620	1620	50	15	[60]
	Hydrogen tank	1	kg	600	600	3	25	[60]
	Fuel Cell	1	kW	370	370	0.019	50 000 (hour)	[61]
	Battery	2	Quantity	218.40	218.40	2.18	–	[56]
	Generator	0.024	Fuel price (€/m ³)	211.57	211.57	0.03 (€/h)	15 000 (hour)	[43]
	Converter	1	kW	375	375	10	25	[59]

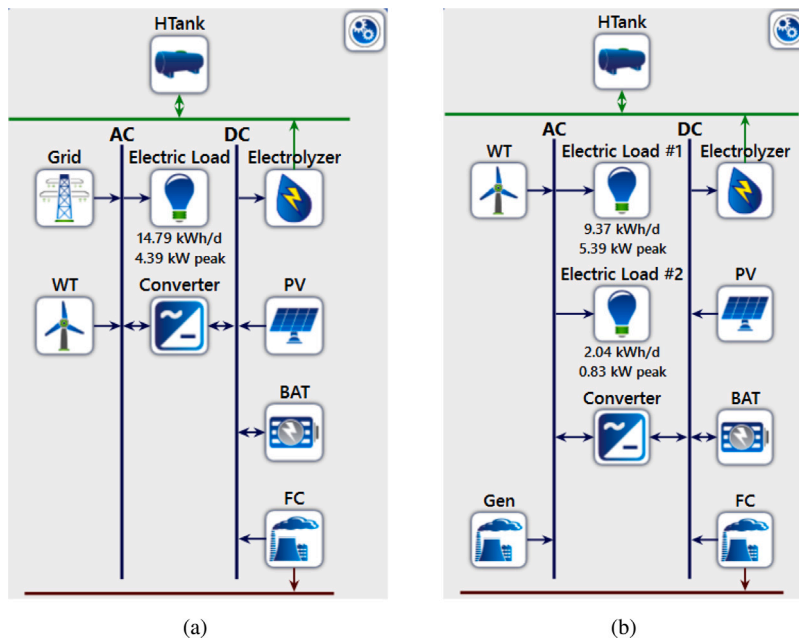


Fig. 8. Modeling in HOMER: (a) BLFARM (b) RIMSAR.

Simulation results

In this section, the techno-economic simulation results for BLFARM and RIMSAR are presented and evaluated. The outcomes from the simulation in 2024 for both farms are first presented. The analysis focuses on the technical and economic aspects of the two configurations, emphasizing their potential for hydrogen production and employment opportunities. Table 3 displays the optimization results for both farms, showcasing three combinations for BLFARM and three for RIMSAR. Specifically, the BLFARM system is analyzed through three scenarios generated by HOMER model simulations: *Scenario 1*, which integrates PV, the main grid, FC and the electrolyzer, and does not include the battery storage system and hydrogen storage tanks; *Scenario 2*, which includes the same components as *Scenario 1* along with batteries; and *Scenario 3*, which includes the same components as *Scenario 1* plus WT. Similarly, the RIMSAR system is evaluated through three off-grid scenarios: *Scenario 4*, which includes PV, FC, electrolyzer and batteries; *Scenario 5*, with the same components as *Scenario 4* and including the LPG; and *Scenario 6*, which eliminates LPG and incorporates a WT. None of these scenarios contain a hydrogen storage tank. As clearly shown in Table 3, Scenario 1 and Scenario 4 emerge as the most optimized configurations, offering the lowest cost and the highest energy production.

For 2024, Fig. 9 illustrates the monthly electricity production for BLFARM and RIMSAR under the best-performing scenarios: *Scenario 1* and *Scenario 4*, respectively. These architectures have the following optimized configurations: *Scenario 1* incorporates a

Table 3
Optimization results for different scenarios.

Farms	System	NPC (€)	LCOE (€/kWh)	Operating cost (€/year)	Initial capital (€)	O&M (€/yr)	Ren Frac (%)
BLFARM	SCENARIO 1	289,308	1.63	9,556	165,776	5,269	83.9
	SCENARIO 2	289,406	1.63	9,554	165,893	5,269	83.9
	SCENARIO 3	289,514	1.5	9,529	166,328	5,224	87.7
RIMSAR	SCENARIO 4	232,662	4.32	5,057	167,282	5,071	100
	SCENARIO 5	234,259	4.35	5,052	168,947	5,087	93.1
	SCENARIO 6	234,773	4.36	5,092	168,951	5,088	100

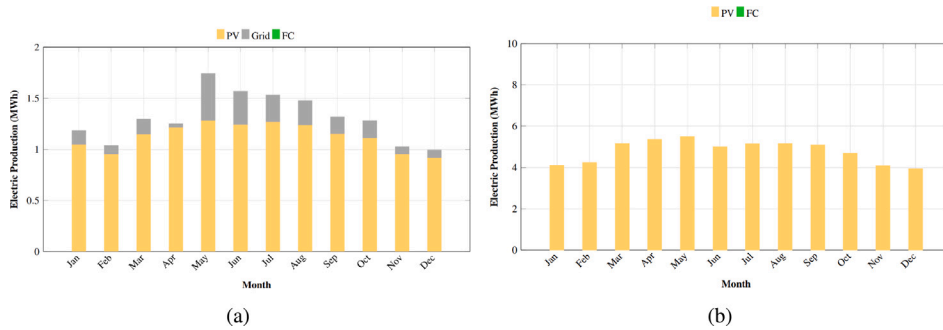


Fig. 9. (a) Electric production for BLFARM in 2024 (b) Electric production for RIMSAR in 2024.

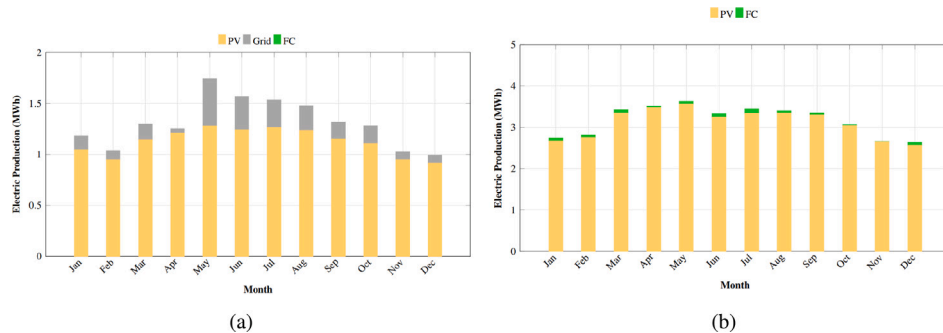


Fig. 10. (a) Electric production for BLFARM in 2050 (b) Electric production for RIMSAR in 2050.

5 kW FC, 18.1 kW PV system, and a 3.20 kW converter. Additionally, it includes a 100 kW electrolyzer. *Scenario 4*, on the other hand, integrates a 5 kW FC, 35.3 kW PV system, 9 kWh battery (BAT), and a 5.92 kW converter. It also includes a 100 kW electrolyzer.

For *Scenario 1*, Fig. 9(a) shows that the PV serves as the primary electricity source throughout the year. The grid supports PV production, especially during the spring and early summer, where grid usage peaks in May and June, and indicates that grid is relied on during higher demands or PV conducted deficits. The contribution of the FC is nonexistent throughout the year, as the reliable grid backup reduces the need for hydrogen-based support.

In *Scenario 4*, Fig. 9(b) shows that the PV systems alone cover all electricity demand all months, while production levels were relatively stable across months. Unlike BLFARM, no grid or generator is included in this scenario. Although the FC is technically integrated into the configuration, it does not provide any electricity in 2024, as the PV system adequately meets the load. This tells us hydrogen is underutilized in this case because of sufficient PV capacity and the absence of extended shortfalls or storage needs triggering FC operation.

In summary, Hydrogen, despite its importance as a key component in the transition to renewable energy systems and its increasing recognition, particularly for applications requiring long-term storage and high energy density, its deployment in the 2024 simulations was not realized. The FC was not activated in either BLFARM or RIMSAR due to low energy demand and the high costs of hydrogen components. These conclusions illustrate the current limitations of hydrogen integration at the individual farm level, yet the system architecture still indicates that there is the potential in the future for energy storage and resilience in energy systems, especially with better economic conditions or at larger community scales.

For the proposed study, to address these challenges, we project system performance for 2050, assuming reduced costs of hydrogen components [64,65]. The 2050 system configurations are based on reduced component costs derived from ongoing technological advancements. The FC in the projected 2050 system has a capital cost of €33.32 per kW, with an identical replacement cost and an

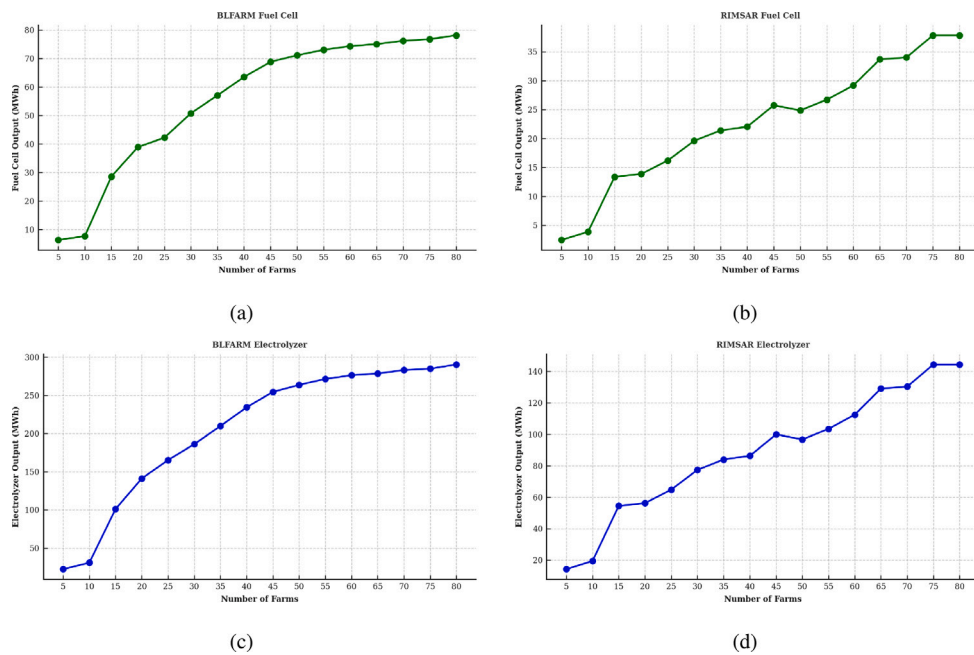


Fig. 11. Annual energy output of hydrogen system components under different community-scale configurations ranging from 5 to 80 farms: (a) FC energy output for BLFARM; (b) FC energy output for RIMSAR; (c) Electrolyzer energy output for BLFARM; (d) Electrolyzer energy output for RIMSAR.

O&M cost of €0.0030 per operational hour. Similarly, the electrolyzer's capital and replacement costs are €130.00 per kW, with an O&M cost of €4.00 per year. Hydrogen storage tanks are optimized at €9.32 per kg for capital and replacement costs, with an O&M cost of €0.080 per year. The lifetime of these components is assumed to be 25 years. As for the other components of the system, they remain the same for this study.

The updated configurations for 2050 are shown in Fig. 10. By 2050, both BLFARM and RIMSAR hybrid systems demonstrated significant cost and efficiency improvements. For BLFARM, the most optimized scenario has PV, FC, electrolyzer and the main grid. The NPC decreased to €28,232, the LCOE to €0.160/kWh, OC to €1,017/yr, initial capital to €15,084, and O&M to €669.45/yr, with the Ren Frac reaching 83.7%. Similarly, RIMSAR saw its NPC drop to €22,905, LCOE to €0.425/kWh, operating costs to €490.74/yr, initial capital to €16,561, and O&M to €489.28/yr, while achieving a Ren Frac of 100%. With the most optimized scenario containing PV, FC, electrolyzer, batteries, electrolyzer and hydrogen storage tank. These results highlight the long-term economic and environmental benefits of hydrogen-integrated microgrids, making them more viable for sustainable agricultural applications.

At the 2050 time frame, advancements in HES technology and cost reductions lead to significant performance improvements for both BLFARM and RIMSAR. For BLFARM, the FC's contribution to electricity output remains the same. As for RIMSAR, FC contribution to electricity output increases slightly.

The transition between 2024 and 2050 highlights the growing role of renewable energy and hydrogen production in farming practices. Although FC use does not vary significantly between the periods, the systems are able to improve their energy autonomy. Minimizing dependence on fossil fuels and promoting long-term sustainability can be achieved with by using excess hydrogen, which can be sold either as electricity or as a fuel in its molecular state, which has the ability to create opportunities for income.

Building on these insights, the analysis was expanded to consider hydrogen integrations at the community level by modeling successively larger microgrids. For both BLFARM and RIMSAR, additional modeling was conducted by scaling the number of aggregated farms in increments, ranging from 5 up to 80 farms.

Fig. 11 presents the results of expanding the modeled microgrid systems from 5 to 80 combined farms. These results illustrate a clear shift in the operational characteristics of hydrogen-related technologies, specifically the FC and the electrolyzer. The findings indicate that both the BLFARM and RIMSAR configurations begin to operate meaningfully around 10 to 15 farms. This range marks a threshold at which excess renewable generation and unmet load conditions emerge, thereby justifying the activation of hydrogen storage and conversion technologies. Subfigures (a) and (b) show the energy output of FCs for BLFARM and RIMSAR, respectively, increases as the number of farms increases. In BLFARM, the annual FC production moves from 6.36 MWh at 5 farms to 78.17 MWh at the maximum of 80 farms, whereas RIMSAR starts from 2.47 MWh at 5 farms and ends at 37.84 MWh at 80 farms. The upward slope confirms both systems are increasingly utilizing the hydrogen pathway for load balancing, particularly beyond 15 farms, where the growing demand begins to justify frequent FC operation. Likewise, subfigures (c) and (d) measure the energy outputs of the electrolyzer. In BLFARM, the output increases from 22.54 MWh to 290.59 MWh, while in RIMSAR it rises from 14.35 MWh to

144.41 MWh. This trend demonstrates that the system is increasingly accommodating the conversion of excess renewable energy into hydrogen, especially in mid-sized facilities. However, after approximately 55 farms for BLFARM and 65 for RIMSAR, the growth rate dramatically decreases. This saturation phase is not a performance limitation or inefficiency, it is a functional and economic optimization to approximately what hydrogen aspects are designed and economically deployed at this stage to guarantee that costly capital infrastructure was fully utilized and not wasted.

The examination of the scaling behavior shows clear differences in operations in both configurations. According to BLFARM, the FC and electrolyzer both have a logarithmically increasing trend: the output increases very steeply between 10 and 45 farms, however beyond 55 farms, the output increases less and less. This explains the diminishing returns phenomena, with hydrogen component input being less relevant as we absorbed the load from other sources, more efficiently. The RIMSAR system was different, with both hydrogen components displaying almost linear scaling until reaching 65 farms, save a mild logarithmic trend beyond this point, as the system begins to reach its internal storage and conversion limits. This delay in saturation gives an idea on reliance on hydrogen in isolated configurations, and also demonstrates greater sensitivity in response to scaled growth at the community level with RIMSAR. As such, the scaling behavior difference between BLFARM and RIMSAR is directly related to the presence or absence of the utility grid, which in BLFARM works as a backup and economic buffer, thereby reducing the hydrogen's importance at higher scalings.

From a techno-economic viewpoint, the saturation is not taking place due to underutilization, it is providing stable system sizing. The hydrogen components contribute to energy security, whilst minimizing excess capital costs, by being close to their utilized limitations without overshoot. This balance achieves NPC and improved operational efficiency without compromising system reliability. From an economic standpoint, scaling from 5 to 80 farms resulted in an increase in all significant cost markers (NPC, operating cost, initial capital and O&M) in a continuous manner relative to the size. This pattern is to be expected in rolling out energy infrastructure for an agricultural community, when more farms are being added to support and supply, towards community-driven farming microgrids [10]. However, the opposite trend was observed for the LCOE, which decreased as the system size increased. For BLFARM, the LCOE reduced from €0.0932/kWh at 5 farms, to €0.0456/kWh at 80 farms; while in RIMSAR, a smaller drop from €0.142/kWh to €0.0743/kWh. The reductions (approximate 51% and 48% reduction in LCOE, respectively), are both examples of economies of scale where fixed and marginal costs are stretched across a larger volume of energy generation. Thus, providing cost-effective and system productive capital at scale.

In summary, the analyses of scale trends, component performance, and economic performance suggest that hydrogen integration, when strategically sized and dispatched, improves both the technical resilience and economic viability of agricultural microgrids. The noted saturation at larger scales for both communities (grid connected and islanded) indicates that there is a mature, optimized system architectures in the development, one where hydrogen is balanced with grid or/and storage resources, the LCOE is minimized, and where over-investment is avoided. The observation supports the viability of community-level hydrogen-enhanced microgrids to sustainably match shares of local, scalable housing and agricultural development with affordable, community-based electricity generation.

It is worth noting that, the successful implementation of hydrogen-integrated hybrid microgrids depends, not only on technical and economic feasibility, but also on social acceptance and supportive policies. Public engagement and financial incentives can accelerate adoption, whereas regulatory inconsistencies may pose challenges. Aligning policy frameworks with community needs is essential for effective integration in agricultural settings. Despite the positive outcomes, the system's performance can be further improved in two critical areas: (i) developing an energy management system to optimize energy production and consumption dynamically and (ii) reducing the costs of HES components, such as FCs, electrolyzers, and hydrogen storage tanks, through technological advancements and economies of scale.

In summary, these findings illustrate the potential of HHMs to transform rural agricultural energy systems with a scalable and sustainable model for future energy demands. Even though hydrogen integration on an individual farm basis has barriers of cost and wasted resources, hydrogen integration at a community level offers a viable approach to technical feasibility and economic viability towards building resilient low-carbon farming communities.

Conclusions and future work

This paper highlights the potential of HHM systems in addressing the energy challenges faced by Moroccan farming systems through a comprehensive techno-economic analysis. By utilizing real meteorological and load data from two pilot farms, a grid-connected BLFARM and an off-grid RIMSAR, the optimal configurations were identified. The 2024 simulations illustrated that both pilot sites use solar PVs as a majority share of the energy source, while the FC does not contribute to total energy output. Scenario analysis identified Scenario 1 (BLFARM) and Scenario 4 (RIMSAR) as the most competitive configurations in 2024, achieving LCOE values of €1.63/kWh and €4.32/kWh, respectively, with renewable fractions of 83.9% and 100%. Despite limited hydrogen utilization due to its high capital costs, these results confirm the technical feasibility of hybrid microgrids for agricultural applications.

Projections for 2050 introduced updated techno-economic assumptions, including substantial reductions in hydrogen component costs. Under these conditions, the BLFARM and RIMSAR systems achieved significant cost improvements, reducing the LCOE to €0.160/kWh and €0.425/kWh, respectively, while maintaining renewable fractions above 83% and 100%. The Net Present Cost (NPC) also decreased from €289,308 to €28,232 for BLFARM, and from €232,662 to €22,905 for RIMSAR, reflecting the impact of technological cost declines and performance gains. However, hydrogen usage remained modest at the single-farm level, indicating limitations in technical and economic feasibility under low-demand conditions.

To address these constraints, the study extended the analysis to community-scale microgrids, aggregating demand across 5 to 80 farms. Results showed strong economies of scale, with the LCOE dropping by over 50% (from €0.093/kWh to €0.045/kWh for BLFARM and from €0.142€/kWh to €0.074/kWh for RIMSAR). The annual FC energy output increased from 6.36 MWh to 78.17 MWh, and electrolyzer output from 22.54 MWh to 290.59 MWh as the number of farms grew, highlighting enhanced utilization of hydrogen infrastructure. These findings demonstrate that community-scale aggregation improves both technical performance and cost efficiency, validating hydrogen's role as an enabler of resilience and energy independence in Moroccan agriculture.

These findings suggest that the viability of hydrogen incorporation in agriculture relies on scale. While individual farms have higher costs and low utilization rates, community-based HHMs unlock both energy autonomy and economic efficiency. The demonstrated scaling effect supports the development of region-wide hydrogen microgrids to serve clustered farming communities. Furthermore, the increase in electrolyzer energy input observed in larger-scale scenarios reflects its use in converting surplus renewable electricity into hydrogen, which can then be stored and used on-site through FCs or sold off-site as a commercial energy product either as electricity or hydrogen fuel, thus creating some new revenue for the farm cooperative.

Future research will focus on validating the irrigation load model against measured pump power using dynamic PV-water-pumping data once available. In parallel, advanced Energy Management System (EMS) integration can be pursued once HOMER extends the MATLAB Link to include hydrogen components. Another promising direction is the development of advanced EMS frameworks in MATLAB/Simulink to complement the techno-economic optimization performed in HOMER. These EMS models would manage power flows at both farm and community scale and incorporate economic exchange mechanisms for cost sharing and energy trading. Furthermore, diversity-aware aggregation methods will be developed to generate realistic community load profiles that can be imported directly into HOMER Pro, reducing repeated runs and better representing non-coincident peak demand.

CRediT authorship contribution statement

S. Chahi: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **A. Aqachtoul:** Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **A. Elamrani:** Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **R. Ouladsine:** Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Supervision. **N. Yousfi Steiner:** Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Supervision. **A. Rochd:** Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition. **M. Bakhouya:** Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mohamed BAKHOUYA reports financial support was provided by MG-FARM. Mohamed BAKHOUYA reports a relationship with International University of Rabat that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by MG-FARM project, Morocco, which is funded by “Ministère de l'Enseignement Supérieur, de la Recherche Scientifique et de l'Innovation, MESRSI” under the LEAP-RE program (Project no: 963530,2022–2026).

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